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Learning to Coordinate through Strategic Game Slices

Final Year Project
Continuous Assessment Report

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Abstract

Negotiation and meeting scheduling involve decisions under private preferences and constraints. We study two capabilities that support coordination in these settings: inferring a partner's relevant hidden information and planning actions from the available evidence. Interaction outcomes alone do not reveal which capability caused a failure. We investigate strategic game slices as a source of targeted supervision. We extend a finite negotiation game with private preferences and investigation, allowing players to acquire information and negotiate binding commitments toward goals with aligned or conflicting interests. A reference solver computes preference posteriors and expected terminal action values. These outputs provide labels for preference inference, direct action selection, and planning conditioned on an oracle preference judgment. We construct slices around selected decisions and use these labels to score generated responses for training a shared model with Group Relative Policy Optimization (GRPO). Diagnostic experiments with Qwen3-4B-Instruct-2507 show that providing a correct preference judgment leaves substantial planning error. In 24 sequential calendar coordination scenarios, the jointly trained checkpoint retains 47 of 72 meetings, compared with 37 for both the base model and complete-game self-play.

1 Introduction

Large language models (LLMs) increasingly participate in multi-agent tasks that require communication, negotiation, and coordinated action. In these settings, an agent's success depends on information distributed across participants and on the decisions those participants make. Interactive benchmarks capture this dependence through collaboration, competition, and decentralised scheduling [1, 2]. However, a failed interaction rarely reveals whether an agent misunderstood its partner or selected an ineffective action despite sufficient understanding. This ambiguity makes both diagnosis and training difficult.

We study two coupled capabilities: *interactive-state inference*, which estimates a partner's task-relevant hidden preferences from available evidence, and *interaction planning*, which uses this information to evaluate and select actions. The distinction follows classical accounts of decision-making with models of other agents [3, 4]. We distinguish the two capabilities experimentally by comparing an agent's actions when supplied with its own preference judgment and with a correct oracle judgment. Errors that remain under the oracle judgment reveal planning difficulties even when the supplied inference is correct.

Our approach uses a finite negotiation game in which players make binding commitments toward goals with aligned or conflicting preferences. Private preferences create uncertainty, while investigation provides an opportunity to acquire information. A reference solver evaluates the consequences of legal decisions, allowing the same environment to support diagnosis and training. We use *game slices*, interaction segments beginning at selected decision contexts, to concentrate supervision on identifiable inference and planning demands. This gives finer control over training content than relying entirely on states visited in complete-game self-play.

The contributions completed at this stage are as follows:

1. We extend a negotiation environment and reference policy evaluation framework that supports matched tests of preference inference, planning, and the value of observed information.
2. We conduct diagnostic and training experiments with Qwen3-4B-Instruct-2507, identifying persistent planning errors under oracle judgments and preliminary transfer gains from slice training.

3. We formulate a slice-selection procedure combining oracle action values, information-omission losses, and model performance gaps to guide coverage of different strategic demands.

2 Literature Review

Inference and planning in interaction. Classical multi-agent decision-making connects opponent models and interactive beliefs to action selection [3, 5]. Bayesian Action Decoder models information conveyed by cooperative actions, while Hierarchical Opponent Modeling and Planning combines goal inference with planning in mixed-motive environments [4, 6]. For LLM agents, explicit mental-state representations support collaboration and hypothesis-based adaptation [7, 8]. However, evaluating theory of mind requires examining adaptation to partners, and strategic-play diagnostics identify gaps between observations, beliefs, and actions [9, 10]. Our diagnostic tests this connection by replacing an explicit preference judgment and measuring the resulting action.

Coordination and negotiation benchmarks. MultiAgentBench evaluates collaboration and competition across interactive scenarios [1]. Commitment-based negotiation exposes long-term consequences of binding agreements [11], while CalBench studies scheduling across private calendars [2]. Dynamic-grounding research further shows that exchanging information does not by itself ensure effective joint planning [12]. These benchmarks motivate controlled games in which inference labels and action consequences can be evaluated separately, followed by transfer testing in a different coordination environment.

Game-based training and curricula. MARSHAL and SPIRAL use strategic self-play to train LLM reasoning, while GIFT studies games as informal training for generalisable abilities [13–15]. Subgame curriculum learning focuses reinforcement learning on selected starting states [16]. Cross-partner coordination work, including Other-Play, examines generalisation beyond self-play conventions [17]. Our approach connects these directions through oracle-labelled game slices: selected decisions support targeted supervision of inference and planning, and evaluation tests whether that supervision transfers to complete interactions. We use grouped relative policy optimisation to learn from the resulting response scores [18].

3 Negotiation Game

We extend the commitment-based negotiation game of Benac et al. [11] with private preferences and investigation. Players use available information to negotiate binding contributions to shared goals.

3.1 Players, Goals, and Utilities

The game contains players $i \in \{0, \ldots, n-1\}$ and commitment variables $x_j \in \{0, 1\}$, each owned by one player. A value of one means that the corresponding contribution is binding. Each goal g

specifies a public set of required commitments R_g and a binary or linear scoring rule:

$$f_g(x) = \prod_{j \in R_g} x_j \quad \text{or} \quad f_g(x) = \frac{1}{|R_g|} \sum_{j \in R_g} x_j. \tag{1}$$

Binary scoring awards one only when every required commitment is present, while linear scoring awards the fraction of requirements met. A commitment can contribute to several goals, so one agreement may affect multiple outcomes.

Player i has preferences $\theta_{ig} \in \{-1, 0, 1\}$, corresponding to avoid, neutral, and want. Its terminal utility is

$$U_i(x, \theta_i) = \sum_g \theta_{ig} f_g(x). \tag{2}$$

Goal progress contributes positively for want, negatively for avoid, and zero for neutral. With no required commitments present, either scoring rule gives zero. An agreement may benefit one player and harm another.

3.2 Negotiation and Private Investigation

A game begins with no binding commitments, no pending offer, and no investigation records. A public finite schedule specifies the proposal opportunities in round-robin order. At each opportunity, the scheduled player chooses OFFER, PASS, or an available INVESTIGATE action.

An offer selects another player and specifies new commitments for the proposer and recipient. It can add commitments for either participant or both, subject to a per-participant addition limit specified by the instance. At least one new commitment must be proposed. The recipient immediately chooses ACCEPT or REJECT. Acceptance makes all proposed additions binding, while rejection leaves the commitment state unchanged. Previously accepted commitments cannot be withdrawn. After the response, play advances to the next scheduled proposal opportunity. The response itself does not consume the recipient's future proposal opportunity.

PASS advances the schedule without changing commitments. INVESTIGATE selects another player's eligible goal preference, consumes one unit of the investigator's query budget, and advances the schedule without changing commitments. The target is publicly observed, while the true preference is revealed only to the investigator. Play ends after the final scheduled opportunity, including the response to any offer made at that opportunity. Utilities are then computed from the final commitments using Eq. 2.

All players observe the goal requirements and scoring rules, proposer schedule, binding commitments, remaining quotas, and public history of offers, responses, passes, and investigation targets. Each player also knows its own complete preferences and retains all answers to its own investigations. Other players' private answers are not directly disclosed. The prior over hidden preferences is public. The recent research candidates use independent uniform draws from want, neutral, and avoid, with each realised preference fixed throughout the game. They exclude queries of publicly disclosed preferences and preferences already directly queried by the investigator. A preference inferred from behaviour remains eligible for investigation.

3.3 Decision Objective and Resource Limits

Players choose actions to maximise their expected terminal utility given their available information, accounting for subsequent negotiation. At ACCEPT/REJECT decisions, an own-utility tie is broken by maximising the sum of other players' expected terminal utilities. If both quantities tie, either response is optimal. Proposal decisions use own utility alone.

The game-design studies add a personal offer quota, consumed immediately when an offer is proposed and never refunded upon rejection. Exhausting this quota prevents the player from making further offers but still allows it to respond to offers from others. Proposal opportunities, offer quota, and investigation budget are separate resources: two round robins give each player two proposal opportunities, while a quota of one permits only one of them to be used for an offer. These opportunity costs do not directly subtract utility.

The recent three-player studies use two round robins, one investigation and one offer per player, and at most one newly added commitment per participant in an offer. Players may own several commitment options and accumulate them through different accepted agreements. The number of options, players, goals, and proposal opportunities is specified by each instance.

Figure 1 illustrates the underlying game before the personal offer-quota extension. The bridge goal uses linear scoring and requires bridge commitments from A, B, and C; the supply goal requires distinct supply commitments from A and C. After C rejects B's bridge offer, A investigates C's supply preference and learns want. B investigates A's bridge preference and also learns want. A and C subsequently agree on supplies, completing that goal at 2/2. B's bridge agreement with A then raises bridge progress to 2/3, while supplies remain at 2/2. Private observations guide partner choice, and agreements accumulate across goals.

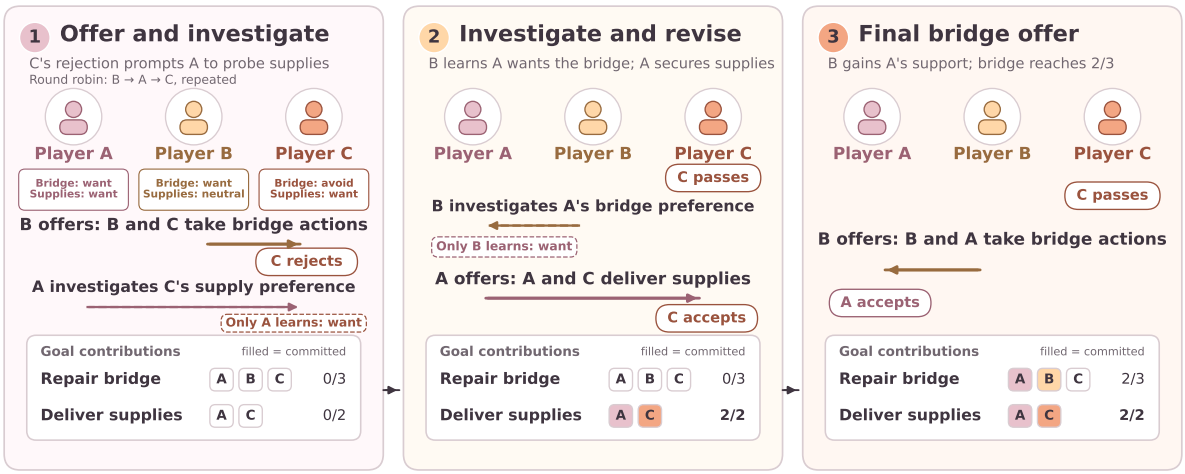


Figure 1: An illustrative three-player trajectory before the personal offer-quota extension. Filled letters indicate binding commitments, and fractions show goal progress. Bridge and supply contributions are distinct commitments. Investigation answers are private to the investigator, and proposal opportunities follow the cyclic order B–A–C.

4 Strategic Game Slices

A strategic game slice begins at a decision context and evaluates a focal player against fixed reference opponents. We distinguish three questions: which actions have high terminal value (Q), how available information supports those decisions (S), and how much value a target model loses (D). The following definitions connect these quantities through a common information structure and continuation model.

4.1 Information Sets and the Reference Oracle

The environment is a finite extensive-form game with hidden preference types [19]. Let θ be the joint type drawn from public prior μ_0, h a public history, and r_i player i's private investigation record. Its decision information is $I_i = (h, \theta_i, r_i)$, together with the public instance. A behavioural strategy $\pi_i(a \mid I_i)$ assigns probabilities to legal actions. Indistinguishable histories share the same strategy. Players retain their observations and actions, giving the perfect-recall structure underlying behavioural-strategy representations of extensive games [20].

The oracle selects a reference profile π^{ref} by searching for mutually consistent best responses over complete contingent strategies. For every player and positive-prior root type, certification requires

$$\max_{\pi_i \in \Pi_i} \mathbb{E}[U_i \mid \theta_i, \pi_i, \pi^{\mathrm{ref}}_{-i}] - \mathbb{E}[U_i \mid \theta_i, \pi^{\mathrm{ref}}] \leq \varepsilon, \tag{3}$$

where Π_i is the set of legal information-respecting strategies. This is a numerical Bayesian Nash best-response condition. Additional checks enforce local optimality of supported actions and the response objective in Section 3, using $\varepsilon = 10^{-8}$ and tie tolerance 10^{-9}. These checks certify the selected numerical reference; sequential equilibrium additionally requires belief consistency at unreached information sets [21].

Behaviour provides evidence through the reference policy. For a candidate world θ compatible with I_i, define its counterfactual prefix weight

$$\lambda_{-i}(h, \theta) = \mu_0(\theta) \prod_{r \prec h:\, j_r \neq i} \pi^{\mathrm{ref}}_{j_r}(a_r \mid I_{j_r}(\theta)). \tag{4}$$

Here $r \prec h$ indexes observed decisions. When the compatible weights have positive total mass, normalising them over worlds consistent with the player's own type and private answers gives its belief $b_i(\theta \mid I_i)$. Own past actions are excluded so that the focal player can deviate from its reference strategy.

4.2 Action Values and Best Responses

After selecting the reference, opponents remain fixed during evaluation. For a focal strategy π_i, write

$$V_I(\pi_i) = \mathbb{E}[U_i \mid b_i(\cdot \mid I), \pi_i, \pi^{\mathrm{ref}}_{-i}]. \tag{5}$$

The action value, optimal value, and regret are

$$Q_i(I,a) = \max_{\pi_i \in \Pi_i:\, \pi_i(I)=a} V_I(\pi_i), \tag{6}$$

$$V_i^*(I) = \max_{a \in \mathcal{L}(I)} Q_i(I,a), \qquad R_i(I,a) = V_i^*(I) - Q_i(I,a). \tag{7}$$

Thus Q forces the current action and optimises the focal player's subsequent contingent behaviour to the native terminal state. It includes later information acquisition, agreements, and responses. With opponents fixed, evaluation becomes an information-constrained best-response problem over the finite tree. Sequence-form representations express strategies through realisation probabilities and flow constraints [22].

4.3 Information Omission and Sensitivity

To evaluate an observed item e, we remove access to one private answer or one completed public proposal event. Rules, own preferences, current commitments, pending offers, quotas, and legal actions remain visible. Contexts with the same retained observations form an omission class $\mathcal{F}_e$. We optimise a shared strategy π^*_{-e} over this entry forest, weighted by normalised counterfactual reach. From entry onward, the strategy retains its remaining information and remembers new actions and observations while the omitted item stays unavailable. Remaining omission-policy ties are uniform after applying the response objective.

Let π_I^* be optimal with all of the player's available observations. The own-utility omission loss is

$$S_e(I) = V_I(\pi_I^*) - V_I(\pi^*_{-e}). \tag{8}$$

Both terms use the same true entry belief and opponents. The omitted strategy is also available to the informed player, so $S_e(I) \geq 0$. Value-of-information methods similarly connect observations to achievable decision quality [23]. Here we evaluate the loss at an already realised context rather than before acquiring the observation.

We distinguish subjective valuation after omission from evaluation at the original context. The subjective table $Q_e^-(\mathcal{F}_e, a)$ averages over the omission class and uses omission-optimal continuation. The table $\widetilde{Q}_e(I,a)$ evaluates that continuation after forcing a at the true entry I. S therefore compares policies under one belief, rather than maxima computed under different beliefs.

Writing $p_e^-(a) = \pi^*_{-e}(a \mid \mathcal{F}_e)$ gives the exact decomposition

$$S_e(I) = S_e^{\text{current}}(I) + S_e^{\text{future}}(I), \tag{9}$$

$$S_e^{\text{current}}(I) = \sum_a p_e^-(a) R_i(I,a), \tag{10}$$

$$S_e^{\text{future}}(I) = \sum_a p_e^-(a)[Q_i(I,a) - \widetilde{Q}_e(I,a)]. \tag{11}$$

The current term evaluates the chosen action with informed optimal continuation, while the future term captures additional loss from continued omission. It follows by substituting $V_I(\pi^*_{-e}) = \sum_a p_e^-(a)\widetilde{Q}_e(I,a)$ into Eq. 8. Consequently, an unchanged current choice can still conceal later information loss.

4.4 Action Profiles and Information Acquisition

A scalar loss does not describe every change in decision structure. We therefore also retain the relative regret vectors $R_i(I,\cdot)$ and $R_e^-(\cdot) = \max_a Q_e^-(a) - Q_e^-(\cdot)$, the optimal action sets, and canonical full/omitted action distributions. The full distribution p^+ is uniform on actions that maximise own utility after applying the response tie rule. Total variation, defined as $\mathrm{TV}(p,q) = \frac{1}{2}\sum_a |p(a) - q(a)|$, and Jensen–Shannon divergence compare it with p_e^-.

These distributions follow the specified tie-breaking convention. Their connection to own-utility loss is

$$S_e^{\text{current}} = (p^+ - p_e^-)\cdot Q_i(I,\cdot) \le [\max_a Q_i(I,a) - \min_a Q_i(I,a)]\ \mathrm{TV}(p^+, p_e^-). \tag{12}$$

Thus switching between equally valuable actions can change the distribution while leaving own loss zero. Relative regret also detects changes among suboptimal actions when the optimal set stays unchanged.

Before investigation, let $\mathcal{L}_0(I_0)$ be the non-investigation actions at entry I_0. The acquisition advantage is

$$G_{j,g}(I_0) = Q_i(I_0, \mathrm{INVESTIGATE}(j,g)) - \max_{a\in\mathcal{L}_0(I_0)} Q_i(I_0,a). \tag{13}$$

This comparison includes the opportunity cost and all subsequent branches.

4.5 Model Difficulty

A slice σ specifies a focal player, an entry distribution ν_σ, and a control/continuation contract. For the new terminal-control design, the model controls the focal player from entry to terminal, while opponents follow π_{-i}^{ref}. If model M induces policy π_i^M, define

$$V_\sigma^* = \mathbb{E}_{I\sim\nu_\sigma}[V_i^*(I)], \tag{14}$$

$$D_\sigma(M) = V_\sigma^* - \mathbb{E}_{I\sim\nu_\sigma}[V_I(\pi_i^M)]. \tag{15}$$

D is the expected terminal performance gap of the actual model under a fixed prompt, tool interface, and decoding protocol. When all N evaluation rollouts complete with terminal utilities $u_i^{(m)}$, its estimate is $\widehat{D}_\sigma = V_\sigma^* - N^{-1}\sum_{m=1}^N u_i^{(m)}$.

Q/S describe the strategic task; D measures the model's difficulty with it. If all responses sampled at a decision receive the same reward, their within-group task advantages are zero even when D is large. We therefore measure sampling diversity and reward contrast alongside D.

4.6 Selecting Slice Entrances

The selection procedure combines strategic coverage from Q/S with measured model difficulty D. Each entrance is a player information context rather than an individual hidden world.

Eligible contexts. We first retain entrances with a certified terminal reference and positive counterfactual reach, then deduplicate identical observation contexts. Different histories or beliefs remain distinct even when commitments and budgets coincide. Each entrance retains its complete visible state, reference identity, and terminal Q/S profile.

Strategic coverage. The profiles identify complementary demands. Positive S marks the value of using an existing answer or historical observation. Its current/future decomposition distinguishes immediate choice from later policy loss. Zero-S contexts can still test valuation when information changes relative regrets while preserving a common optimal action, or planning when a myopic action is inferior under terminal Q. Information-acquisition candidates are checked for both the query's Q advantage and the value of receiving its answer compared with taking the same public query action without answer access. This separates acquisition from public signalling.

Diversity and model difficulty. Candidate selection will cover different parents, goal structures, decision phases, and remaining focal-decision lengths, protecting acquisition–use chains without discarding useful single-decision tasks. Concentration by parent and physical structure will be limited while preserving distinct information contexts. We will construct 800 candidates including a reserved validation portion. Under a fixed prompt, decoding protocol, and focal control to terminal, repeated model rollouts on the training candidates will estimate D and sampled action/reward diversity. These measurements will guide selection of 100 training slices within the coverage design, rather than ranking all entrances by S, Q span, or distribution distance.

5 Model Diagnosis

We separate inference from planning at an explicit judgment–action interface. Inference maps observed evidence to a judgment about a partner's hidden preference; planning maps the current decision state and a supplied judgment to an action. Independent calls let us replace the inference output while holding the planning task fixed.

5.1 Task Separation and Oracle Labels

The diagnostic contains 22 two-player cases across seven game structures. Each queries one hidden partner–goal preference $z \in \{\text{want}, \text{neutral}, \text{avoid}\}$. The evidence comprises six direct investigation answers, six single-choice elimination cases, six ambiguous single-choice cases, and four multi-step updates. For case c, the reference oracle conditions on the public prior, own preferences, private answers, and observed partner choices to obtain $p_c^*(z)$.

The inference label $J_c^* = (\mathcal{S}_c^*, m_c^*)$ contains the support $\mathcal{S}_c^* = \{z : p_c^*(z) > 0\}$ and a favoured preference m_c^*. The latter is the highest-probability value only if its probability exceeds the runner-up by more than 0.1; otherwise, it is undetermined. The model receives the original visible evidence and submits the corresponding support and favoured value through a belief-submission tool. These labels supervise what the evidence supports, rather than requiring the model to guess the realised hidden value.

Planning receives the rules, goal requirements, known preferences, current commitments, pending offer, schedule position, remaining investigation budget, and legal action tools. Behavioural history and the queried private answer are replaced by the supplied qualitative judgment. Neither numerical posterior probabilities nor oracle action values are disclosed. The model must select an action using this judgment and the retained physical state.

The direct-action call measures performance without an explicit supplied inference. For the

Table 1: Four diagnostic calls at each matched decision. Each call is a fresh request. The two planning calls differ only in the supplied judgment.

Condition	Input	Scored output
Inference	Original state and visible evidence	Preference judgment $\widehat{J}_c$
Direct action	Original state and visible evidence	Native game action
Plan: model	History-free state and $\widehat{J}_c$	Native game action
Plan: oracle	Same history-free state and J_c^*	Native game action

paired planning calls in Table 1, tools, decoding settings, and request seed are identical. Removing behavioural evidence prevents planning from simply reconstructing the original inference, and replacing only the judgment measures its effect on the selected action.

5.2 Supervision and Protocol Validation

Exact rational checks verify that all posteriors compatible with the oracle judgment share the same optimal-action set. The scorer's tolerance-based acceptable set is checked over these regions as well. We also recompute legal actions and continuation payoffs from the history-free state to verify that removing history preserves the planning problem.

Inference correctness requires both submitted label components to match J_c^*. Action correctness uses the certified acceptable-action set, with an own-utility tolerance of 0.1 and the diagnostic's response social-tie tolerance of 0.1. These are historical scoring tolerances, separate from the reference solver's numerical tolerance. The same label format is used for the inference and conditional-planning training objectives described in Section 6.

To measure consequences, let $W_c(a, z)$ be the diagnostic's fixed per-preference terminal continuation payoff for action a, and define

$$v_c(a) = \sum_z p_c^*(z) W_c(a, z), \tag{16}$$

$$\mathrm{Reg}_c(a) = \max_b v_c(b) - v_c(a), \tag{17}$$

$$\Delta_c = v_c\big(\widehat{a}_c(J_c^*)\big) - v_c\big(\widehat{a}_c(\widehat{J}_c)\big). \tag{18}$$

Here $\widehat{a}_c(J)$ denotes the model's action given judgment J. Both actions are evaluated under the same oracle posterior and the diagnostic's fixed reference continuation, rather than rescored under their different supplied judgments. Positive Δ_c indicates that judgment replacement improves the decision.

The suite includes nine repair-sensitive cases, where an alternative preference can require an incompatible acceptable action, and thirteen action controls. All four multi-step cases are controls with a common acceptable action across preferences. This distinction identifies cases where judgment replacement can require an action change.

5.3 Diagnostic Results

We evaluate Qwen3-4B-Instruct-2507 [24] once per case at temperature zero, with a 4,096-token output limit. One model-judgment planning response has a format failure. It counts as incorrect for accuracy and is excluded from paired utility comparisons, leaving 21 valid pairs.

Table 2: Base-model diagnostic accuracy. Joint inference requires both posterior support and the most likely preference to match the oracle.

Evidence	Inference	Direct	Plan: model	Plan: oracle
Direct investigation answer	6/6	2/6	3/6	3/6
Single-choice elimination	0/6	3/6	2/6	3/6
Ambiguous single choice	5/6	2/6	2/6	2/6
Multi-step update	0/4	4/4	4/4	3/4
Total	11/22	11/22	11/22	11/22

Table 2 reveals distinct inference and planning failures. The model recovers all six directly revealed preferences, yet planning with oracle judgments succeeds in only three of these six cases. Across all cases, each action condition scores 11/22. Among ten valid pairs with incorrect initial judgments, replacement improves utility in three, leaves it unchanged in six, and reduces it in one. Across all 21 valid pairs, mean regret falls from 0.684 to 0.541 despite unchanged aggregate accuracy. Supplying a correct inference can help, but does not reliably resolve the remaining planning difficulty.

6 Training and Testing

The diagnostic results motivate training preference inference and conditional planning alongside direct action selection. We evaluate the trained checkpoints on private-calendar coordination, which uses different actions and interaction mechanics from the negotiation game.

6.1 Training Objectives and Comparisons

All training variants use Qwen3-4B-Instruct-2507 as their base model. In direct-action training, oracle labels score the model's proposed actions. The action-and-planning variant additionally trains decisions conditioned on oracle preference judgments, and joint training adds explicit preference inference. These tasks update a shared model through grouped reinforcement learning [18]. At test time, the model selects actions from its observations without receiving oracle judgments.

Complete-game self-play provides a comparison in which players learn from their own terminal utilities through full interactions. We also evaluate the unchanged base model. Table 3 denotes direct-action, action-and-planning, and joint training by GameSlice-A, GameSlice-AP, and GameSlice-Joint, respectively.

6.2 Transfer to Sequential Calendar Coordination

The preliminary transfer experiment uses a local sequential adaptation of CalBench [2]. Four agents coordinate three meetings in each of 24 fixed scenarios. Agents observe private calendars and communicate to establish common meeting slots while preserving earlier commitments. Every player in a team uses the same checkpoint. We report retained meetings out of 72 and complete games out of 24. Excess cost is computed by subtracting the minimum feasible team cost from the realised cost, dividing by four, and averaging over complete games. Communication and burden imbalance are averaged over agents and games. Communication counts messages per round-level scheduling success, with a minimum denominator of one; imbalance measures deviations in excess

Table 3: CalBench transfer on 24 frozen sequential games. Meetings count final retained meetings; excess cost is per agent and conditional on full-game success. Communication and imbalance average native per-agent values over all games.

Model	Meetings ↑	Full games ↑	Cost ↓	Comm. ↓	Imbalance ↓
Baseline	37	5	0.100	5.663	0.443
Self-play	37	6	0.625	5.181	0.755
GameSlice-A	38	8	0.688	5.003	0.490
GameSlice-AP	44	10	0.075	4.760	0.432
GameSlice-Joint	47	9	0.194	4.714	0.292
Qwen3-4B (thinking)	42	7	0.571	4.828	0.557
MARSHAL Generalist	37	6	1.333	4.429	0.875
Social-R1-4B	39	5	1.100	4.554	0.901

rescheduling burdens. We additionally compare Qwen3-4B (thinking), MARSHAL Generalist, and Social-R1-4B [25].

GameSlice-Joint retains ten more meetings than the baseline and self-play and has lower communication overhead and burden imbalance. GameSlice-AP completes the most full games and has the lowest conditional excess cost (Table 3). Cost averages cover different successful subsets for different models. These preliminary comparisons motivate further testing of conditional-planning supervision for transfer. Since training exposure and inference settings differ across the runs, the next phase will compare matched budgets over repeated runs.

7 Discussion

Beyond preference inference. Our games locate uncertainty primarily in partners' preferences. Real coordination also requires reasoning about what others have observed and what they believe others know, as explored in recent multi-agent theory-of-mind work [8, 26, 27]. These hidden states belong within inference, but introduce a further challenge: the same refusal may reflect conflicting preferences or missing information, calling for different responses. Future game slices could vary knowledge and preferences independently, testing whether agents distinguish these causes and choose when to clarify or disclose information. The key question is which distinctions in a partner's mental state are necessary for a good decision.

Coverage and interaction. Game slices let us select strategically consequential decisions and verify their supervision. Self-play provides coherent interaction trajectories, but visits states according to the current policies and may encounter important decisions infrequently. This motivates targeting such decisions explicitly, consistent with work on subgame curricula [16]. A possible next step is to combine the two: use self-play to identify recurring failures, construct and solve slices around those decisions, and return to complete games to evaluate the learned behaviour. This would test whether targeted practice can repair local weaknesses while preserving effective coordination over an entire interaction.

Inference under unfamiliar partner policies. Our oracle assumes a fixed partner policy, while self-play primarily exposes a model to copies of its evolving policy. Yet the meaning of an observed action depends on how the partner chooses: a rejection that rules out a preference under one policy may be compatible with it under another. Cross-partner coordination research

motivates training with diverse partner populations [17, 28]. For game slices, this means varying planning horizons, tie-breaking rules, or action noise and recomputing labels under each behavioural model. Evaluation on held-out policy families would test whether agents can interpret unfamiliar behaviour and adapt their plans beyond the assumptions of a single oracle.

8 Research Plan for the Next Semester

Building on the selection procedure in Section 4.6 and the discussion of partner generalisation, next semester's work will compare prompt systems with post-training and identify the capabilities responsible for improved coordination.

8.1 Prompt Systems versus Post-Training

We will compare modifying the inference-time prompt system with updating model parameters through slice-based post-training. The prompt system will structure evidence tracking, judgments about partners, and action planning while keeping model weights fixed. A factorial comparison will evaluate four conditions: the base model with the original prompt, the base model with the structured prompt system, the post-trained model with the original prompt, and the post-trained model with the structured prompt system. All conditions will receive the same task information and action tools.

This design tests whether prompting and post-training provide independent, complementary, or substitutable benefits. We will compare negotiation regret and calendar coordination at matched inference budgets, recording generated tokens, model calls, and latency. Training cost will be reported separately to assess the trade-off between one-time parameter updates and repeated inference overhead. Applying the same prompt system to both checkpoints will also test whether training improves the use of explicit reasoning support. Prompt variants will be selected on validation tasks before evaluation on the test set.

8.2 Which Capabilities Improve and Transfer?

We will examine improvements in four capabilities: maintaining and updating beliefs from interaction history; identifying useful information to seek or disclose; planning actions using inferred constraints; and revising plans while preserving commitments across participants and turns. Matched decision probes will compare base and post-trained models on the same visible contexts. The inference/planning intervention in Section 5 will test whether gains remain when both models receive the same correct judgment. Training-objective ablations will examine the contributions of inference and conditional-planning supervision.

In calendar coordination, we will track whether agents incorporate new constraints, check proposed times with affected participants, and repair conflicts while preserving existing meetings. Controlled scenarios and corrected-judgment interventions will link these process measures to final meeting retention and cost, connecting decision-level gains to complete coordination outcomes.

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